به نام خدا

استاد: خانم رضایی | آزمایشگاه سیستم دیجیتال ۲ | علی سپهرنیا (۹۶۱۲۱۰۳۸)

پاسخ تمرین جلسه پنجم

برنامه ای بنویسید که دمای اندازه گیری شده توسط LM۳۵ را در دو عدد نمایشگرسون سگمنت نشان دهد.

G:\Personal\Uni\Micro Lab\Micro\Az5-Tamrin\Az5-Tamrin.c

Notes | Az5-Tamrin.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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Project : Session5-Exercise
Version :
Date    : 10/27/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>
#include <delay.h>

// Declare your global variables here
int i,digit[2],sevenseg[] = {0x3f, 0x06, 0x5b, 0x4f, 0x66, 0x6d, 0x7d, 0x07, 0x7f, 0x6f};
float temp;

// Voltage Reference: Int., cap. on AREF
#define ADC_VREF_TYPE ((1<<REFS1) | (1<<REFS0) | (1<<ADLAR))

// Read the 8 most significant bits
// of the AD conversion result
```

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```
unsigned char read_adc(unsigned char adc_input)
{
ADMUX=adc_input | ADC_VREF_TYPE;
// Delay needed for the stabilization of the ADC input voltage
delay_us(10);
// Start the AD conversion
ADCSRA|=(1<<ADSC);
// Wait for the AD conversion to complete
while ((ADCSRA & (1<<ADIF))==0);
ADCSRA|=(1<<ADIF);
return ADCH;
}
void DigitSeperation(int i)
{
    digit[0] = i % 10;
    digit[1] = (i / 10) % 10;
    digit[2] = (i / 100) % 10;
}
void SevenSegment()
{
    //digit0
    PORTC = 0b00001110;
    PORTD = sevenseg[digit[0]];
    delay_ms(1);

    //digit1
    if(temp>9 || temp<-9) //If Number Is 2digt Turn On Its 7-Segment
    {
        PORTC = 0b00001101;
        PORTD = sevenseg[digit[1]];
        delay_ms(1);
    }

    //digit2
    if(temp>99 || temp<-99) //If Number Is 2digt Turn On Its 7-Segment
    {
```

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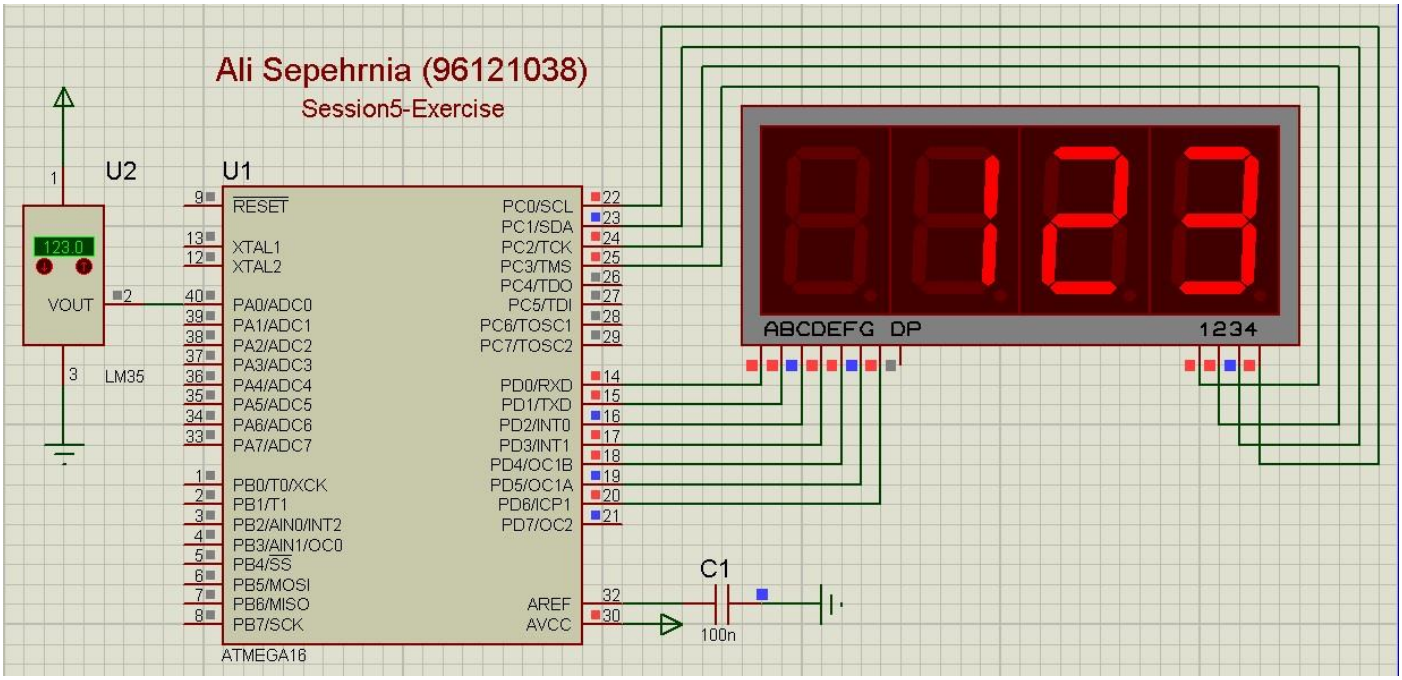
Notes | Az5-Tamrin.c

```
72          PORTC = 0b00001011;
73          PORTD = sevenseg[digit[2]];
74          delay_ms(1);
75      }
76  }
77  void main(void)
78  {
79  // Declare your local variables here
80  float Dvoltage,Avoltage;
81
82  // Input/Output Ports initialization
83  // Port A initialization
84  // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
85  DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
86  // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
87  PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);
88
89  // Port B initialization
90  // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
91  DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (0<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
92  // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
93  PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);
94
95  // Port C initialization
96  // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=Out Bit2=Out Bit1=Out Bit0=Out
97  DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (1<<DDC3) | (1<<DDC2) | (1<<DDC1) | (1<<DDC0);
98  // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=0 Bit2=0 Bit1=0 Bit0=0
99  PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);
100
101 // Port D initialization
102 // Function: Bit7=Out Bit6=Out Bit5=Out Bit4=Out Bit3=Out Bit2=Out Bit1=Out Bit0=Out
103 DDRD=(1<<DDD7) | (1<<DDD6) | (1<<DDD5) | (1<<DDD4) | (1<<DDD3) | (1<<DDD2) | (1<<DDD1) | (1<<DDD0);
104 // State: Bit7=0 Bit6=0 Bit5=0 Bit4=0 Bit3=0 Bit2=0 Bit1=0 Bit0=0
105 PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);
106
107 // ADC initialization
```

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```
107 // ADC initialization
108 // ADC Clock frequency: 500.000 kHz
109 // ADC Voltage Reference: Int., cap. on AREF
110 // ADC Auto Trigger Source: ADC Stopped
111 // Only the 8 most significant bits of
112 // the AD conversion result are used
113 ADMUX=ADC_VREF_TYPE;
114 ADCSRA=(1<<ADEN) | (0<<ADSC) | (0<<ADATE) | (0<<ADIF) | (0<<ADIE) | (0<<ADPS2) | (0<<ADPS1) | (1<<ADPS0);
115 SFIOR=(0<<ADTS2) | (0<<ADTS1) | (0<<ADTS0);
116
117 while (1)
118     {
119     // Place your code here
120     Dvoltage=read_adc(0);
121     Avoltage=Dvoltage*2.56/256;
122     temp=Avoltage*100;
123     DigitSeperation(temp*1.0001); //To prevent c from rounding down false
124         for(i=0;i<10;i++) //To make display longer
125         {
126           SevenSegment();
127         }
128     }
129 }
130
```


Ali Sepehrnia (96121038)
Session5-Exercise
U1
U2
ATMEGA16
LM35
VOUT
123.0
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
C1
100n
ABCDEFG DP
1234